

AI Executive Academy
Capstone Project

AI and the Future of Retail Banking

Project Organisation & Four-Session Workshop Plan

Team of 5 · 4 Workshops × 2 Hours · 7-Minute Final Presentation

EXECUTIVE SUMMARY

Retail banking stands at one of the most consequential inflection points in its history. AI is not a marginal productivity tool here — it is restructuring how deposits are gathered, credit is underwritten, payments are routed, and wealth is advised. The institutions that move with speed and strategic clarity will widen their moat decisively. Those that move slowly will face margin compression from both sides: AI-native fintechs attacking from below, and large platform players (Big Tech) attacking from above.

This document organises your team's four workshop sessions into a rigorous, MIT-quality analytical process. Each session has a clear purpose, a set of hands-on AI activities, and individual member tasks designed to maximise learning while driving toward an excellent capstone result.

PROJECT STRUCTURE

The Four Banking Areas Under Analysis

Each of the four core retail banking domains carries distinct AI disruption dynamics. Your team will cover all four:

Banking Areas

- Deposits & Savings — AI-driven personalisation of savings products, predictive churn, conversational banking
- Lending & Credit — AI underwriting, alternative data scoring, instant decisioning, fraud detection
- Payments & Transactions — Real-time fraud scoring, smart routing, embedded payments, AML/KYC automation
- Wealth Management — Robo-advisory, hyper-personalisation, AI portfolio rebalancing, digital financial planning

Capstone Deliverables

- Part 1: 6–8 slide industry analysis (landscape, opportunities, threats, gap, vision, recommendations, investment thesis)
- Part 2: 2–4 slide behind-the-scenes (AI tool usage, critiques, team reflection)
- 7-minute live presentation on final day
- Working AI agent prototype to demonstrate during Q&A

Strategic Thesis (Proposed)

Core Argument

AI is bifurcating retail banking into two groups: AI-native challengers with near-zero marginal cost of service, and legacy incumbents whose cost structures, culture, and regulatory posture are creating a dangerous deployment gap. The winners will be those who close the gap fastest — not by automating existing processes, but by reimagining the banking relationship from first principles.

TEAM ORGANISATION & MEMBER ROLES

Each team member owns one banking domain plus a cross-cutting strategic or production role. This structure ensures deep domain expertise while covering all presentation components. The AI Lead role is the most hands-on technically and is recommended for the team member most interested in building agents.

Member A [AI Lead + Deposits & Savings]

Focus Area: Deposits, Savings, Conversational Banking

- SESSION 1: Build the shared prompt library in Notion/Google Doc. Run the first AI research sprint on deposits/savings using Claude. Generate a 'who wins, who loses' analysis for your domain.
- SESSION 2: Lead the agent-building exercise. Set up a multi-step Claude agent: (1) search for recent news, (2) summarise key findings, (3) extract strategic implications. This agent becomes the team's demo piece.
- SESSION 3: Use AI to generate and stress-test the investment thesis. Run a bull/bear case prompt. Document 2+ examples of AI outputs that needed heavy human revision for the 'Behind the Scenes' slides.
- SESSION 4: Lead Q&A prep — use AI to generate 10 tough board-level questions. Run the red-team prompt: 'What is weakest about this retail banking AI analysis?'
- INDIVIDUAL: Research Nubank, Monzo, and Starling as AI-native deposit competitors. Use AI to generate a comparison table across 6 dimensions.

Member B [Lending & Credit Lead + Deck Narrative]

Focus Area: Lending, Credit, Underwriting, Fraud

- SESSION 1: Run structured AI research on AI underwriting and alternative credit scoring. Use Perplexity to find 3 real case studies (JPMorgan Chase COiN, Upstart, Zest AI). Bring findings to share-out.
- SESSION 2: Lead the 'deployment gap' exercise for lending. Build a gap matrix using Claude: map each lending use case by AI feasibility vs. current adoption. Rate barriers (regulatory, cultural, technical, org).
- SESSION 3: Draft slides 1–3 (landscape, opportunities, threats) using AI. Use a design AI tool (Gamma.app or Beautiful.ai) to generate initial slide layouts. Iterate with the team on narrative arc.
- SESSION 4: Own the final deck narrative — make sure the story flows across all slides. Use AI to rewrite any slide that feels disconnected. Present the lending section in the final rehearsal.
- INDIVIDUAL: Deep-dive on the regulatory barrier to AI in lending (Fair Credit Reporting Act, CFPB guidance on explainability). Use AI to summarise the key constraints in plain language.

Member C [Payments & Transactions Lead + Visualisations]

Focus Area: Payments, Fraud Detection, AML/KYC, Embedded Finance

- SESSION 1: Research AI in payments: real-time fraud scoring, smart routing, embedded finance. Focus on Visa/Mastercard AI, Stripe Radar, and open banking. Use Claude to map the payments value chain.
- SESSION 2: Build the 'deployment gap' matrix for payments. Use AI to generate a visual framework (table or 2x2 matrix: impact vs. adoption) for where AI is deployed vs. where it should be.
- SESSION 3: Own all data visualisations in the deck. Use Claude + code execution (or Datawrapper/Flourish) to generate charts: AI adoption rates, market sizing, fraud loss reduction data. Every chart needs a cited source.
- SESSION 4: Present the payments section in rehearsals. Own the visual consistency of the final deck (fonts, colors, chart styles). Export final PDF.

- **INDIVIDUAL:** Research the EU AI Act implications for payments AI and KYC automation. Use AI to draft a 1-paragraph plain-English summary for the 'threats' slide.

Member D [Wealth Management Lead + AI-Native Vision]

Focus Area: Wealth Management, Robo-Advisory, Personalisation

- **SESSION 1:** Research AI in wealth: robo-advisors (Betterment, Wealthfront, Schwab Intelligent Portfolios), hyper-personalisation, AI financial planning. Use AI to identify which wealth models are most threatened.
- **SESSION 2:** Lead the 'AI-native bank design' workshop. Facilitate the team brainstorm using a structured AI prompt: 'Design a retail bank with AI at its core optimised for maximum customer benefit.' Capture outputs.
- **SESSION 3:** Draft the 'Vision' and 'Recommendations' slides (slides 5–6). Use AI to generate the recommendations, then apply your judgment to sharpen them. Each recommendation needs a named incumbent or market signal.
- **SESSION 4:** Present the wealth and vision sections in rehearsals. Lead the 'Team Reflection' portion of the Behind the Scenes slides.
- **INDIVIDUAL:** Compare three AI-native wealth models (Betterment, Wealthfront, a traditional private bank). Use AI to generate a comparison framework and then critique the AI output yourself.

Member E [Investment Thesis Lead + Behind the Scenes]

Focus Area: Cross-Cutting: Investment Signals, AI Tools Documentation

- **SESSION 1:** Take the 'AI landscape' shared role. Use AI to scan recent analyst reports (Goldman Sachs, McKinsey, Accenture) on AI in banking. Build a market signals briefing for the team's opening slide.
- **SESSION 2:** Begin investment thesis research in parallel. Use AI to identify 8–10 asset candidates: AI-native banks, incumbent banks with strong AI programmes, AI infrastructure plays, and fintech ETFs.
- **SESSION 3:** Lead the investment thesis deep dive. Use Claude to generate bull and bear cases for each asset. Force-rank top 4–5 picks with rationale. Draft the 'Investment Thesis' slide.
- **SESSION 4:** Own the 'Behind the Scenes' section entirely. Document which AI tools were used and how. Identify 2+ examples where AI output was misleading or required major revision. Draft the team reflection narrative.
- **INDIVIDUAL:** Research the market reaction to AI announcements by major banks (JPMorgan, Goldman, BBVA). Use AI to generate a timeline of key AI investments in retail banking since 2020.

FOUR-SESSION WORKSHOP PLAN

S1
FOUNDATION
Framing the Landscape + AI Research Sprint

Purpose: Orient the full team, divide the work, and run the first structured AI research pass across all four banking areas. Leave with a shared findings document and a draft thesis.

Agenda (2 hours)

- 00:00 — Capstone brief walkthrough & alignment (15 min)
- 00:15 — Map retail banking: 4 areas, value chain, AI entry points (20 min)
- 00:35 — Assign roles & domains; confirm shared tools (Google Doc, Notion, shared prompt library) (10 min)
- 00:45 — Parallel AI research sprint: each member researches their domain (35 min)
- 01:20 — Share-out: top 3 findings per area, identify cross-cutting themes (20 min)
- 01:40 — Agree on guiding thesis, key questions, and session 2 prep tasks (20 min)

Hands-on AI Activities

- Set up shared prompt library — everyone contributes and reuses prompts
- Run 'who wins, who loses' prompt for each banking area
- Use Perplexity or Claude web search to find 2–3 real case studies per domain
- Generate a 'AI value chain map' for retail banking using Claude
- Produce first-draft competitive threat matrix (incumbents vs. challengers vs. Big Tech)

Session 1 Output

Shared findings document with AI opportunities + threats across all 4 banking areas. Draft thesis statement. Prompt library started.

S2
ANALYSIS
Deep Dive: Agents, Deployment Gap & AI-Native Vision

Purpose: Go beyond research into structured analysis. Build and demonstrate a working AI agent. Quantify the deployment gap. Design the AI-native bank from scratch.

Agenda (2 hours)

- 00:00 — Review individual prep and findings (20 min)
- 00:20 — Agent build: Member A leads setup of a multi-step Claude research agent (30 min)
- 00:50 — Deployment gap exercise: estimate feasibility vs. adoption per use case, per area (25 min)
- 01:15 — 'Build from scratch' workshop: design the AI-native retail bank as a team (25 min)
- 01:40 — Synthesise: identify 4–5 biggest strategic insights for the presentation (20 min)

Hands-on AI Activities

- Build a multi-step agent: research → summarise → extract strategic implications (demo piece)
- Generate a deployment gap matrix (use case × feasibility × adoption × barrier type)
- Run a structured 'greenfield bank design' prompt, then use AI to critique and stress-test the output
- Use AI to draft slide outlines for all 8 Part 1 slides
- Ask AI to identify the 3 biggest internal contradictions in your analysis so far

Session 2 Output

Deployment gap matrix. AI-native bank concept. 5 strategic insights. Working agent prototype ready to demo.

S3**BUILD****Presentation Build + Investment Thesis**

Purpose: Assemble the full deck from drafted components. Develop the investment thesis with rigour. Build the 'Behind the Scenes' documentation. First complete run-through.

Agenda (2 hours)

- 00:00 — Review all individual draft slides; align on narrative arc (25 min)
- 00:25 — Investment thesis deep dive: assets, bull/bear cases, rationale (30 min)
- 00:55 — Fill evidence gaps: data charts, market sizing, cited sources (25 min)
- 01:20 — Draft 'Behind the Scenes' slides with the full team (20 min)
- 01:40 — First full timed run-through of complete deck (20 min)

Hands-on AI Activities

- Use AI to generate bull + bear investment cases for each asset pick
- Use Gamma.app, Beautiful.ai, or Canva AI for slide design polish
- Generate data visualisations using Claude code execution or Datawrapper
- Ask AI to 'steelman the weakest argument in this presentation'
- Document all AI outputs requiring significant human revision (required for Part 2 slides)

Session 3 Output

Complete draft deck (Part 1 + Part 2). Investment thesis with ranked assets and rationale. AI tool usage log documented.

S4**FINAL PREP****Rehearsal, Polish & Delivery Prep**

Purpose: Tighten every slide, rehearse the 7-minute delivery twice, prepare for hard Q&A, and export the final deck. Leave with full confidence in the presentation.

Agenda (2 hours)

- 00:00 — Incorporate all individual revisions; final deck review (20 min)
- 00:20 — Full rehearsal #1 — timed, no stopping (10 min)
- 00:30 — Hot debrief: cut anything over 7 min, sharpen weak slides (20 min)
- 00:50 — Q&A prep: generate + practice 10 tough questions (25 min)
- 01:15 — Full rehearsal #2 — polished run (10 min)
- 01:25 — Final polish, export, speaker assignments (25 min)

Hands-on AI Activities

- Use AI to generate 10 board-level Q&A questions
- Run red-team prompt: 'What is most vulnerable to challenge in this analysis?'
- Use AI to sharpen executive summary language and recommendation wording
- Generate speaker notes for each slide using AI as a starting point
- Finalise 'Behind the Scenes' reflection on what AI got wrong and what surprised the team

Session 4 Output

Final polished deck (PDF + editable). Speaker assignments. Q&A prep sheet. Team has rehearsed the 7-minute delivery at least twice.

INDIVIDUAL WORK PLAN (BETWEEN SESSIONS)

The sessions are intensive but bounded. The quality of the final presentation depends heavily on the individual preparation done between sessions. Below is the recommended between-session work plan for each member.

Period	All Members	Domain Lead Task	AI Learning Goal
After S1	Read one AI-in-banking report. Add 2 case studies to shared doc.	Write 3–5 bullet points on the deployment gap in your area.	Practice writing a multi-turn prompt that iterates on an initial AI answer.
After S2	Draft your 1–2 slides using AI. Run them through at least one AI critique loop.	Research 1–2 competitors or case studies that make your section argument sharper.	Experiment with prompting Claude to play 'devil's advocate' on your section.
After S3	Refine your slides based on run-through feedback. Practice your 60–90 second section.	Verify all data points and citations in your section. Replace any unsourced AI claims.	Document one specific moment where AI produced something you had to significantly rewrite.
Final Day	Review speaker notes. Arrive with backup PDF. Be ready to demo the agent.	Own your Q&A question area. Know your domain cold.	Reflect: what would you do differently with AI tools on the next project?

RECOMMENDED AI TOOLS BY TASK

Every team member should develop fluency with at least 3–4 of the tools below. The goal is not just to use AI — it is to use it visibly, critically, and with documented judgment about where it helped and where it fell short.

Tool	Best For	Recommended Use in This Capstone
Claude (Anthropic)	Deep analysis, long-form synthesis, agent building	Primary research engine. Build the research agent in Session 2. Run all multi-turn analytical prompts.
Perplexity AI	Cited web research, market signals	Grounding analysis in real headlines and data. Finding recent analyst reports and news.
ChatGPT / GPT-4o	Code generation, data analysis, image creation	Generating Python/Excel visualisations. Creating charts from data. Slide image generation.
Gamma.app / Beautiful.ai	AI-powered slide design	Rapidly generating slide layouts from bullet points. Especially useful for Session 3 deck build.
Canva AI	Visual design, infographic creation	Creating framework visuals (2×2 matrices, deployment gap diagrams, value chain maps).
Notebook LM (Google)	Research synthesis from uploaded documents	Uploading analyst reports and bank annual reports, then asking cross-document questions.
Datawrapper / Flourish	Data visualisation	Creating publication-quality charts from banking data with proper citations.

QUALITY STANDARDS FOR AN EXCELLENT RESULT

A strong MIT-level capstone in this domain meets all of the following criteria. Use this as your team's internal quality checklist before the final presentation:

- Grounded in evidence: every claim backed by a named company, data point, or cited report. No generic 'AI will transform banking' statements without specifics.
- Demonstrates AI fluency visibly: the audience can see that AI was used throughout — not just to write text, but to research, analyse, visualise, build, and critique.
- Takes a clear point of view: the presentation makes an argument, not just a survey. You have a thesis and defend it.
- Deployment gap is quantified: your gap analysis is specific. Not 'AI is underused' but 'AI could reduce mortgage underwriting time from 3 weeks to 24 hours; adoption is currently under 15% of US institutions.'
- Investment thesis is rigorous: each asset pick has a named thesis (not just 'AI will grow'). There is a bull case, a bear case, and a reason you chose this asset over the obvious alternatives.

- Behind the Scenes is honest: the best teams are candid about where AI fell short, where they had to heavily revise outputs, and what they learned about AI's current limitations as a research tool.
- Presentation discipline: 7 minutes means 7 minutes. Rehearse ruthlessly. Every slide should be able to stand alone as a single defensible claim.

A NOTE ON APPROACH

The best AI-assisted analysis is not AI-generated analysis. It is human judgment, sharpened and accelerated by AI. Your job is not to prompt Claude and paste the output. Your job is to use AI as a thinking partner — to pressure-test your hypotheses, generate hypotheses you would not have considered, surface data you would not have found, and iterate faster than was ever possible before.

The deployment gap between what is possible and what is being done in banking is real — and your team should embody the closing of that gap in how you work. Use AI ambitiously. Revise it critically. Present the result with confidence.

Good luck — the best capstone teams do not just present the future of banking. They demonstrate it.